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Micromachines (Basel). 2022 Mar 26;13(4):520. doi:
[10.3390/mi13040520](https://doi.org/10.3390/mi13040520)

Review of Ultrasonic Ranging Methods and Their Current Challenges

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PMCID: PMC9025471 PMID: [35457823](#)

Abstract

Ultrasonic ranging has been widely used in automobiles, unmanned aerial vehicles (UAVs), robots and other fields. With the appearance of micromachined ultrasonic transducers (MUTs), the application of ultrasonic ranging technology presents a more extensive trend. This review focuses on ultrasonic ranging technology and its development history and future trend. Going through the state-of-the-art ultrasonic ranging methods, this paper covers the principles of each method, the signal processing methodologies, the overall system performance as well as key

ultrasonic transducer parameters. Moreover, the error sources and compensation methods of ultrasonic ranging systems are discussed. This review aims to give an overview of the ultrasonic ranging technology including its current development and challenges.

Keywords: ultrasonic ranging, transducer, pulse echo, time of flight, error compensation

1. Introduction

An ultrasonic transducer is a device that can realize the mutual conversion between high-frequency electrical energy and mechanical energy. It is generally divided into the piezoelectric type, capacitive type and magnetoelastic type and is widely used in measurement fields, including distance measurement (in the air) [1], sonar [2], medical imaging [3,4], ultrasonic therapy [5], non-destructive testing [6,7], flow monitoring [8,9], etc.

Ultrasonic ranging has the advantages of relatively low hardware requirements over radio frequency ranging and laser ranging in short-range measurement due to ultrasound's slower transmission speed. Besides, ultrasonic is not sensitive to ambient light, electromagnetic interference, dust or other factors. In addition, the relatively low cost of ultrasonic transducers is user-friendly for engineering applications such as reversing radar and construction surveys. Therefore, ultrasonic ranging is suitable for small range (the distance is usually less than 10 m) and high-precision (the accuracy can reach the level of millimeter generally) non-contact distance measurement, and the derived two-dimensional positioning [10] and three-dimensional positioning [11,12], object shape recognition [13] and multi-sensor fusion trajectory measurement [14], etc.

Ultrasonic ranging originated with underwater sonar in the 1940s, and as it was developed for use in the air in the 1960s, possibilities for contactless distance measurement from 0.2 m to 2 m in the air opened up [15]. In the 1980s, the research and application of ultrasonic ranging systems gradually began and was

first applied to robot obstacle avoidance [[16,17,18,19](#)]. Since the 1990s, the research on ultrasonic ranging methods has gradually deepened, and not only does the traditional time of flight (ToF) method continue to be studied [[20,21](#)], but two frequency continuous wave (TFCW) method [[22](#)], multi-frequency continuous waves (MFCW) method [[23](#)], binary frequency shift keying (BFSK) method [[24](#)] and amplitude modulation (AM) method [[25](#)] have also been proposed, different new measurement methods continue to emerge. Since the beginning of this century, MEMS ultrasonic devices have been gradually applied to ultrasonic ranging, and different signal processing methods have been gradually developed including thresholding method, curve-fitting method, sliding-window method and optimum correlation detection in [[26](#)]. Different coding methods have also been gradually applied, developing ultrasonic ranging systems towards miniaturization, high precision and high processing speed. At present, in addition to improving the performance of ultrasonic devices and ultrasonic systems, there is also optimization of the processing algorithm aimed at the high performance and wide applications of ultrasonic ranging systems.

Nowadays, ultrasonic ranging is widely used in the automotive industry, UAVs, robots, and industrial auxiliary measurement equipment. In the automotive industry, it is mainly used for the detection of obstacles to assist the driving system with achieving anti-collision function, including distance measurement to obstacles [[27,28](#)] and acquisition of kinematics information for obstacles moving around the vehicle [[29](#)]. Additionally, the type of road surface can be identified through reflected signals [[30](#)] to assist in safe driving. Similarly, ultrasonic sensors can also be applied to obstacle avoidance of UAVs [[31,32](#)] and robots [[33](#)]. In addition, it can also be used to detect the landing state [[34](#)] to ensure the flight safety of UAVs. In the field of robots, indoor positioning is also a common application. Whether position coordinate measurements [[35](#)] or attitude measurement [[36](#)], it is realized through the fusion of multiple transmitters in the space and multiple receivers on the robot, and the positioning accuracy can reach the level of centimeters [[37](#)]. Meanwhile, the auxiliary

navigation system can realize the path planning of the robot [38]. In the industrial field, ultrasonic equipment can be used to monitor the field environment such as the tower crane in [39] to ensure the normal operation of industrial equipment.

Conventional bulk piezoelectric ultrasonic transducers are not practical in mobile applications due to their large size and high power consumption compared to micromachined ultrasonic transducers (MUTs) [40]. MUTs can be integrated into portable products, such as smartphones, wearable devices, etc. As a result of research and technical development, the performance of MUTs has been improved, and they are hoped to replace the bulk piezo-composite ultrasonic transducers. For example, multiple ultrasonic sensors can be integrated into mobile devices for distance estimation, which could then be combined with other information of reflected ultrasonic signals to realize gesture recognition and classification [41,42], to achieve contactless human–computer interaction. In addition, a sliding-window based method [43] can be used to improve accuracy. Therefore, MUTs can be applied to smartphones, VR devices and smart home products.

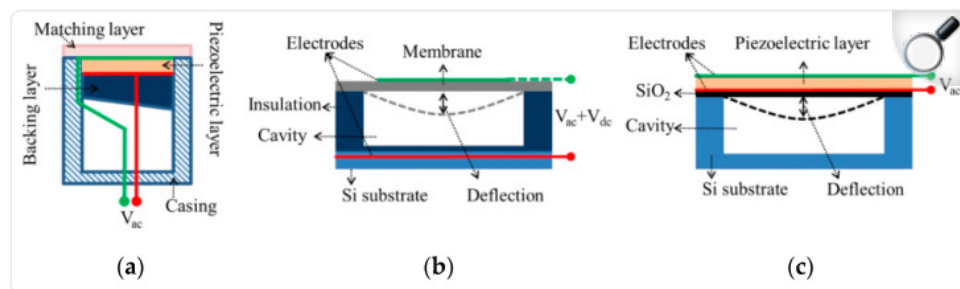
In this study, the status of ultrasonic ranging is reviewed from different aspects. [Section 2](#) introduces the characteristic parameters of ultrasonic transducer related to ranging application, including frequency and impedance, energy conversion characteristics, and ultrasonic propagation characteristics, and also analyzes the influence on ranging systems. [Section 3](#) presents the principle, composition and evaluation parameters of ultrasonic ranging systems. [Section 4](#) summarizes the various measurement methods and signal processing methods, as well as the corresponding measurement range, measurement accuracy and measurement rate. [Section 5](#) focuses on the error source and error compensation method of ultrasonic ranging. In the final section, conclusions are drawn, existing problems are summarized, and predictions of the future development trend are provided.

2. Transducer and Ultrasonic Characteristics Related to Ranging

2.1. Comparison of Different Transducers

An ultrasonic transducer is a kind of energy conversion device, which converts mechanical energy to electrical energy or vice versa. Although conventional bulk piezoelectric transducers can generate high output power, acoustic impedance mismatches between the devices and transmission medium severely reduce transduction efficiency [44]. A typical structure of a conventional piezoelectric ultrasonic transducer is shown in [Figure 1a](#). In order to reduce the mismatch of acoustic impedance between the piezoelectric layer and the propagation medium, a matching layer usually is added at the transducer's front face [45,46,47,48,49]. For the transducers operating with an air load, the best properties of a matching layer are observed from the materials of polyether sulfone and nylon membranes [47]. A matching layer based on a combination of a porous material with a low-density rubber material can achieve an improvement in received signal amplitude of 30 dB when compared with the unmatched case [48]. In addition, transducers based on ferroelectrets have been reported with reduced impedance mismatch with the air [50].

Figure 1.



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Typical cross-sectional structures of (a) piezoelectric ultrasonic transducers; (b) CMUTs and (c) d31-mode PMUTs [51].

In contrast to conventional ultrasonic transducers, the MUTs employ a flexural membrane for generating and receiving ultrasound waves, which is conducive to better impedance matching and higher transmitting efficiency [51]. Therefore, MUTs are mainly divided into capacitive micromachined ultrasonic transducers (CMUTs) [52] and piezoelectric micromachined ultrasound transducers (PMUTs) [53]. CMUTs' operation is based on the flexural vibrations caused by a field-induced electrostatic attraction between the suspended membrane and the substrate (Figure 1b), whilst PMUTs is based on flexural vibrations caused by d31- or d33-mode excitation of a piezoelectric membrane (Figure 1c).

CMUTs have the advantage of high bandwidth (often over 100%) [51], which is more conducive to the modulation of the ranging system. However, the output pressure of CMUTs depends on the excitation voltage and the inverse of the capacitor gap [54]. Hence, the CMUTs usually have submicrometer gaps [55] which leads to not only a complicated fabrication process but also a small linear vibration amplitude. To overcome this problem, high bias voltages of hundreds of volts can be used [56], which, however, would lead to further increase in the system complexity and higher power consumption.

Compared with CMUTs, PMUTs do not require DC bias voltage [57,58], and the linear displacement range is a function of the membrane thickness [59]. For PMUTs, the active piezoelectric layer, e.g., lead zirconate titanate (PZT) and aluminum nitride (AlN), is deposited with nanofabrication techniques, e.g., plasma-enhanced chemical vapor deposition (PECVD), sol-gel process, and sputtering, on passive elastic layers such as Si, Si_xN_y or SiO_2 . The resonant frequency of the PMUTs does not directly depend on the thickness of the piezoelectric layer. Instead, the flexural mode resonant frequencies are closely related to the shape, dimensions, boundary conditions, intrinsic stress and mechanical stiffness of membranes [53]. In practical applications, PMUTs can generate sufficient ultrasonic power from a sub-mW electrical drive signal for target ranges up to a few meters [60] and can meet the

requirements of output pressure in the range. Therefore, PMUTs are expected to be a better solution for rangefinders [61] due to their superior power efficiency even though that is less than conventional bulk transducers. As an example, a power dissipation of 400 μ W at 30 fps for a 1 m maximum range has been achieved with an AlN-based ultrasonic array transducer [62]. The comparisons of bulk piezoelectric transducers, CMUTs and PMUTs are shown in Table 1.

Table 1.

The characteristics of bulk piezoelectric transducers, CMUTs and PMUTs [45,51,52,53,54,56,57,58,59].

	Bulk Piezoelectric	CMUTs	PMUTs
Fabrication methods	Mechanical machining, e.g., dicing, lapping and casting	Wafer bonding and micromachining	Micromachining and wafer transfer diaphragm formation
Matching layer	Required	No matching layer	No matching layer
k_{eff}^2 * and bandwidth	Low	High	Low
CMOS compatible and flip-chip integration	No	Yes	Yes
DC bias requirement	No	Yes	No
		Millimeter level size of	Millimeter level size of

General Size	Centimeter to millimeter level size	transducer arrays Hundred microns level size of a single CMUT	transducer arrays Hundred microns level size of a single PMUT
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* k_{eff}^2 : electromechanical coupling coefficient.

2.2. Transducer Characteristics

The following subsections describe some characteristics of transducers related to the ultrasonic ranging system’s performance.

2.2.1. Frequency Characteristics

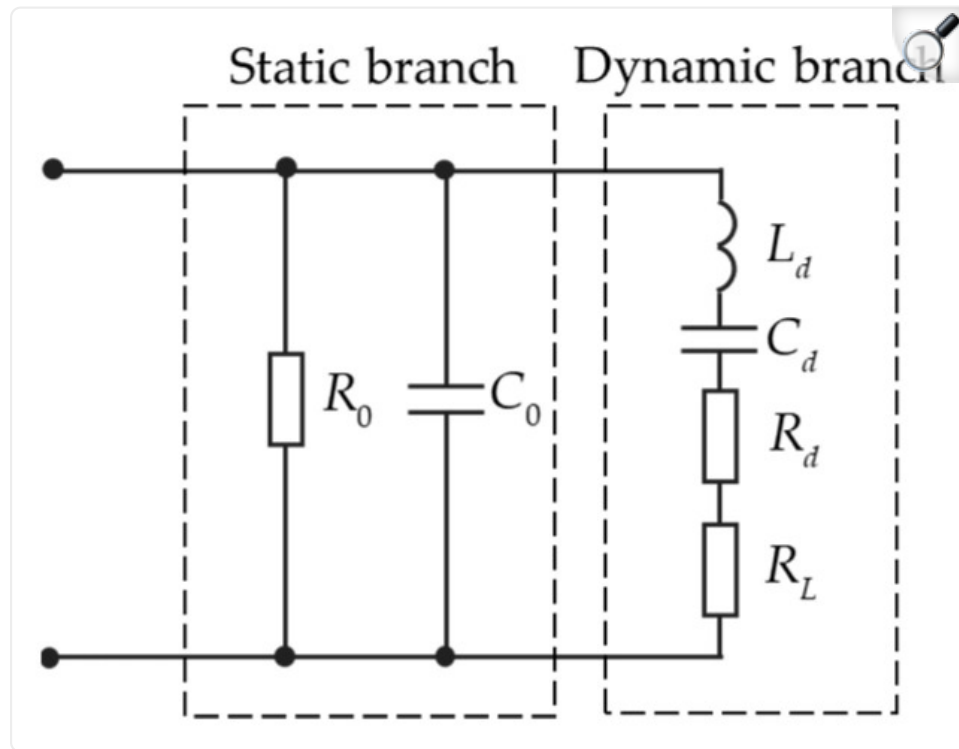
The operating frequency of the ultrasonic ranging transducer is often chosen to be near the series resonant frequency which has minimal impedance. The frequency bandwidth of PMUTs used for ranging in the air is generally narrow, e.g., 25% in [63], compared with CMUTs, >100% reported in [61]. Current ranging systems generally use a certain frequency of the transducer rather than frequency modulation. The double resonant frequency transducer [64] is also under study, which can obtain a broadened frequency response. Broader bandwidth allows the adoption of the frequency modulation operation and is conducive to the improvement of ranging technology. In addition, the operating frequency determines ranging accuracy. In general, the higher the working frequency, the higher the ranging accuracy. However, this will be at the expense of the ranging range as the attenuation of ultrasound in the propagation medium will increase proportionally to the frequency [65]. Therefore, the choice of operating frequency should be matched with specific

measurement requirements.

2.2.2. Impedance

Impedance analysis of transducers can evaluate the dynamic characteristics of transducers [66]. At present, most ultrasonic transducers are made of polycrystalline piezoelectric ceramic materials. High-frequency electrical signals, >20 kHz, are applied to piezoelectric materials and converted into ultrasonic signals. The equivalent circuit model of the piezoelectric ultrasonic transducer is shown in [Figure 2](#).

Figure 2.



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Equivalent circuit model of transducer.

Where R_0 is the parallel resistance of medium loss, C_0 is the

static capacitance, measured far below the resonant frequency, C_d is dynamic capacitance, L_d is dynamic inductance, R_L is the load resistance. In general, $R_0 \gg R_d$. In order to simplify the model, R_0 can be ignored during analysis. In addition, R_L is short-circuited during analysis. Then, the equivalent impedance of the piezoelectric transducer can be expressed as Equation (1) [67]

$$Z = \frac{(R_d + j\omega L_d + 1/j\omega C_d) \cdot 1/j\omega C_0}{R_d + j\omega L_d + 1/j\omega C_d + 1/j\omega C_0} \quad (1)$$

To maximize the power transmission efficiency of the ranging system, make the dynamic branch in resonance, and the resonant frequency $\omega = \omega_s = 1/\sqrt{L_d C_d}$. Then the impedance of the transducer can be shown as Equation (2), as a combination of both resistance and reactance.

$$Z = R + jX = \frac{R_d}{1 + (\omega_s C_0 R_d)^2} - j \frac{\omega_s C_0 R_d^2}{1 + (\omega_s C_0 R_d)^2} \quad (2)$$

Matching circuits are commonly included in the ultrasonic system, aiming to make the transducer closer to pure resistance in order to achieve higher power output and better efficiency [68,69]. An improvement of 300% in the amplitude of the received signal was reported with and without the matching circuit [68]. This improvement leads to an improved signal to noise ratio (SNR) and, therefore, better accuracy in measurements which is desirable in the ultrasonic ranging system.

2.2.3. Electromechanical Coupling Coefficient

The electromechanical coupling coefficient k_{eff}^2 is a parameter of ultrasound transducers to reflect energy conversion efficiency. A higher k_{eff}^2 is desired to have more energy to be converted between electrical and mechanical energy within the transducer [4]. The k_{eff}^2 can be calculated using Equation (3) [70]

$$k_{eff}^2 = 1 - \left(\frac{f_r}{f_a} \right)^2 \quad (3)$$

where f_r is the resonance frequency and f_a is the anti-resonance frequency.

In general, the k_{eff}^2 of PMUTs used is less than 5%, and the method called ClarinetTM to increase the coefficient is also being studied [71]. Due to the high attenuation in ultrasound waves along the travelled path in the air, the ultrasonic ranging system would benefit from a transducer with a higher coupling coefficient to improve its signal sensitivity and measurement accuracy.

Apart from optimizing for a higher coupling coefficient of the transducer, other approaches are used to improve the SNR of the received signals as well for the ranging system. In [72], the authors reported that the signal amplitude of the bent transducer was enhanced 10 times bigger by designing its active diameter. The signal quality can also be improved by optimizing the excitation voltage of the transmitting transducer [73], exciting the transmitter transducer with a direct modulation system [74] and adopting the broadband tuning of the receiving circuit [75].

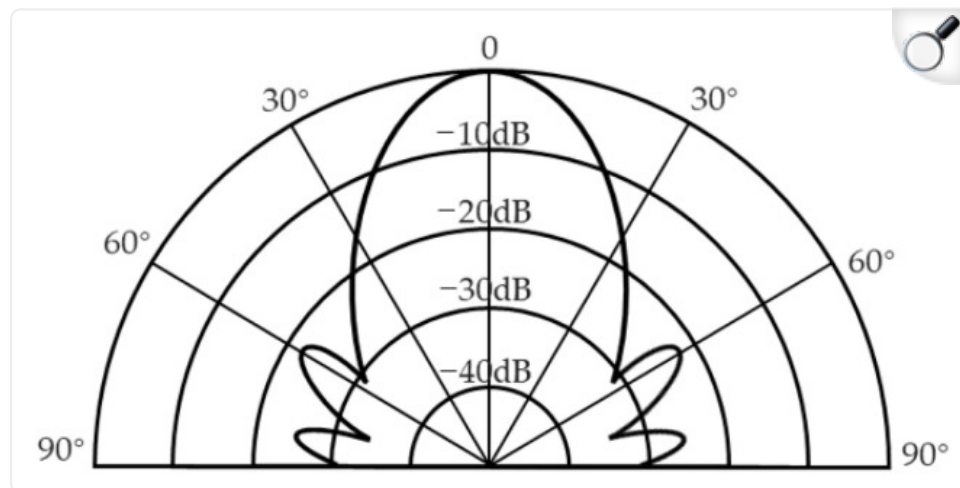
2.2.4. Directivity

The directivity of transducer and transducer array is a characteristic that the amplitude of transmitting response or receiving response changes with azimuth angle. It determines the range of the azimuth angle of the measurable range in the space. Depending on their applications, the requirements for directional characteristics of transducers are different. For a transmitting transducer, the sharpness of its directional characteristic curve determines the concentration of its transmitted energy, and for a receiving transducer, it determines the range of azimuth to explore space. In the ultrasonic ranging system, the transducers can be used as either the transmitter, the receiver or both depending on the configurations.

The sound field established by the transducer in the medium is related to the shape and size of the transducer, the vibration mode, the working parameters such as frequency and the types of the medium [76].

The directivity response diagram can be obtained by drawing the directivity response diagram in decibel (dB) compared to its maximum in the sound field. [Figure 3](#) presents the directivity diagram of a transmitting ultrasonic transducer.

Figure 3.



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Typical directivity diagram of a transmitting ultrasonic transducer.

As shown in [Figure 3](#), there are the main lobe and sidelobes in the acoustic field [77], with the main lobe mainly could be used for ranging. In applications where the ranging direction is fixed, the transducer is generally configured in an orientation to align with its main lobe's direction.

When the width of the main lobe becomes narrower, the energy of

the beam is more concentrated and its directivity is higher. The parameter, beam spread angle θ , can be used to represent the directivity of the ultrasound transducer. Beam spread angle measures the width of the beam in degrees, from side to side of the main lobe where the ultrasonic energy intensity drops to -3 dB. In general, the ranging system takes the corresponding space within the beam spread angle of the transducer as the detectable range.

When the beam spread angle of a single-element ultrasonic transducer cannot meet the range requirement, transducer array and beam forming techniques can be adapted to change the directivity of transducers [78] and inhibit the sidelobe level [79], and can also improve the ultrasonic emission intensity and increase the effective measurable distance.

2.3. Ultrasonic Propagation Characteristics

When the ultrasonic wave propagates in the medium, its sound pressure intensity gradually decreases, known as attenuating. Attenuation is a result of several factors from interaction with medium and interfaces, including scattering, absorption, reflection and diffraction. In the ultrasonic ranging system where the transducer operates in pulse-echo mode, the transducer emits the ultrasonic wave and is reflected by the target obstacle, the ultrasonic overall loss formula can be expressed as shown in Equation (4), including the attenuation contribution from the propagation path [76]

$$G = \frac{P_r}{P_t} = G_{ac} \frac{a}{4r} 10^{-2\alpha r} \quad (4)$$

where G is ultrasonic attenuation coefficient, P_r is the pressure of the ultrasonic wave received by the transducer, P_t is the pressure of the ultrasonic wave emitted by the transducer, G_{ac} is the acoustic gain, depends on the size of the target, a is effective membrane radius, r is the range to the target, α is attenuation

coefficient which increases with the frequency of sound waves, and also depends on humidity, temperature and ambient pressure. The ultrasound attenuation in the air can be as high as 7 dB/m at 215 kHz at room temperature and 60% relative humidity in the air [63]. In general, the ultrasonic ranging system tends to use transducers that operate at frequency range < 300 kHz, and typically at frequencies around 40 kHz, in order to obtain a balance between the measurement distance of interest, the measurement resolution and the signal quality, e.g., SNR.

3. Ultrasonic Ranging System and Its Evaluation Parameters

The ultrasonic transducer is the key component of the ultrasonic ranging system. The system measures the ultrasonic wave transmitted and received by the transducer and converts it into distance measurement to complete the function of the ranging system.

3.1. Principle of Ultrasonic Ranging

The ultrasonic transducer transmits ultrasonic waves and receives echo reflected from the object by either the same transducer, pulse-echo mode or by another transducer as pitch-catch mode. When using the pulse-echo mode, the distance s between the ultrasonic transducer and the object is to be calculated by measuring the time t between the transmitted signal and received signal as shown in Equation (5)

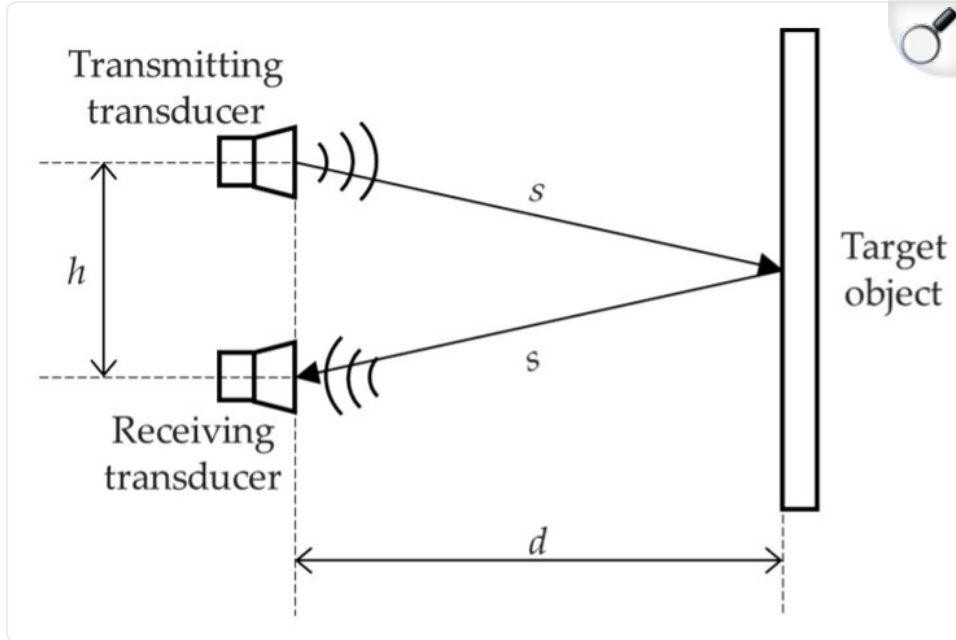
$$s = ct/2 \quad (5)$$

where c is the sound velocity in the medium.

[Figure 4](#) shows a pair of the transducers are configured in pitch-catch configuration, the distance between the plane where ultrasonic transducers are located and the target object d is

$$d = \sqrt{s^2 - (h/2)^2} \quad (6)$$

Figure 4.



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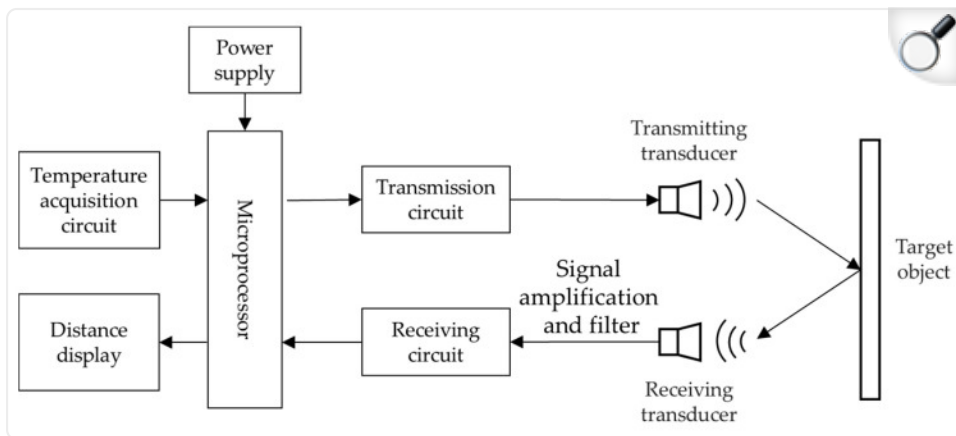
Schematic diagram of Ultrasonic Ranging with a pair of transducers.

In practice, d is much greater than the distance between the two transducers h and therefore h can be treated as negligible. That means $d = s$ and Equation (6) can also apply in this configuration.

3.2. Composition of Ultrasonic Ranging System

A typical ultrasonic ranging system would be as shown in [Figure 5](#) and composed of transducers for transmitting and receiving, transmitting circuits, receiving circuits, a microprocessor, a temperature compensation module and a display module [\[80,81,82,83\]](#).

Figure 5.



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An overall diagram of ultrasonic ranging system.

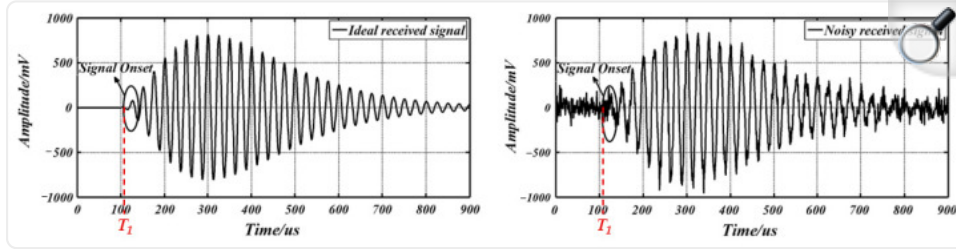
Transducers include the transmitting transducer and the receiving transducer. The transmitting circuit amplifies the programmed pulses to high voltage pulses that can drive the transducer, while the preamplifier of the receiving circuit amplifies the received signals because they are generally weak [84]. Then, the signal passes through the bandpass filter to remove interfering noise signals and improve the SNR of received signals. To avoid the high requirements of the hardware, the bandpass sampling theory can be applied to digitalize the echo signal [85].

Microprocessor generally adopts a micro controller unit (MCU) which is high-performance and low-power [86,87,88]. The microprocessor controls the programming of the transmitting signal. At the same time, it also processes the received signal to obtain the information of interests and transmits the measurement results to the display module. Meanwhile, the power supply module is the basis of the normal work of the ranging system.

As for the processing of received signals by the microprocessor,

the key is to obtain the starting time T_1 of echo signals, the ideal and noisy ultrasonic echo signals are shown in [Figure 6](#). In addition, the amplitude of the received signal varies with the measured distance [\[89\]](#). Different methods can be used to obtain T_1 , which will be elaborated in detail in [Section 4](#).

Figure 6.



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Ideal and noisy ultrasonic echo signals [\[90\]](#).

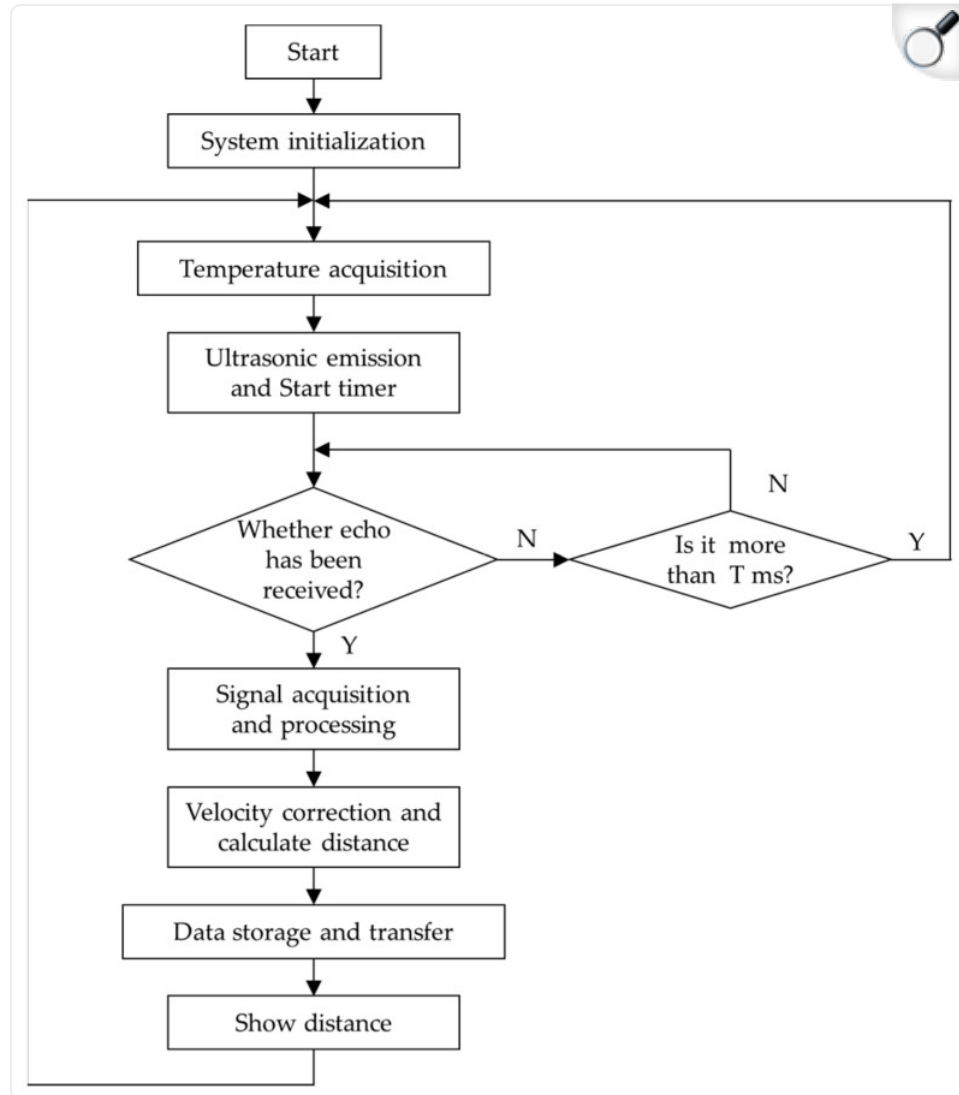
The temperature compensation module compensates for the influence of temperature on the propagation velocity of ultrasonic waves [\[80\]](#). For every 1 °C increase in temperature, the speed of sound increases by about 0.607 m/s [\[91\]](#). The circuit is equipped with a temperature sensor and connected with the microprocessor to realize the acquisition of real-time ultrasonic velocity.

Display module: For the measurement data obtained by the microprocessor, they can be directly connected to a liquid crystal display (LCD) for real-time display [\[92\]](#) or transmitted to other microcontrollers through wireless communication to avoid the inconvenience of reading the measurement data directly [\[93\]](#).

Several research groups had developed their own ultrasonic ranging system [\[80,82,94,95,96\]](#), and we extract and summarize the program flow of the ultrasonic ranging system and show in [Figure 7](#). The value of T is set in direct proportion to the furthest

measurement distance, that is, the single measurement period is inversely proportional to the value of T .

Figure 7.



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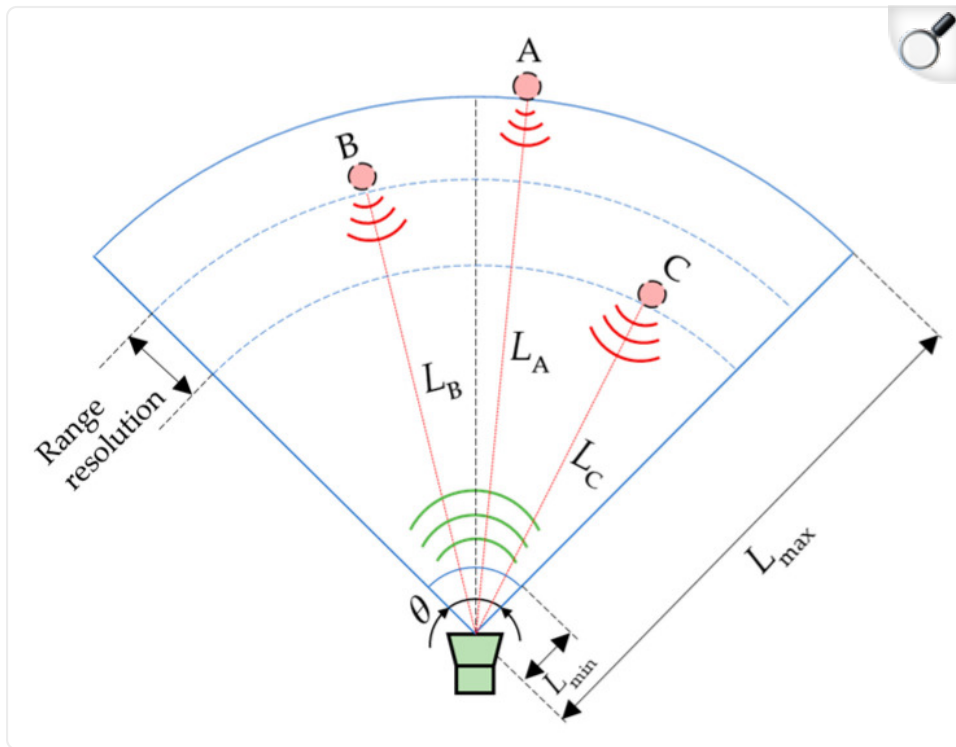
A program flow of an ultrasonic ranging system.

3.3. Evaluation Parameters of Ranging System

3.3.1. Measurement Range of Distances and Angles

The sector in a two-dimensional space or the conical shape in a three-dimensional space can be represented as the effective range of ultrasonic ranging as shown in [Figure 8](#) [97]. In the figure, θ is the beam spread angle which relates to the directivity of the transducer; $L_{\min}$ and $L_{\max}$ are the nearest and the farthest distance that can be measured, respectively. $L_{\min}$ is determined by the dead zone of the transducer, which is related to the near field of the ultrasound propagation. The dead zone is also related to the length of the excitation pulses [98]. Although the longer the excitation signal is, the more energy delivered and therefore the better signal quality at reception, its longer ringing down time will lead to a larger blind area. $L_{\max}$ is determined by the propagation attenuation characteristics of the ultrasonic wave described in [Section 2.2.4](#). The measurement range of an ultrasonic ranging sensor is 2 cm–5 m in general to ensure adequate echo signal at reception [99].

Figure 8.



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Schematic diagram of measuring range.

In order to reduce the dead zone effect on the measurable range, the pitch-catch mode can be adopted with separate transmitters and receivers [100]. In this configuration, the pulse's ringing down will not mask the echo received, therefore, reducing the dead zone $L_{\min}$. Another approach is to optimize the transmit pulse and receiving filter by customizing the transmit pulses of chirp 80–65 kHz with Cauer high-pass filter to reduce the receive pulse duration and combining with partial ringing removal by using ringing look-up-tables to reduce the minimum detectable range down to 3 cm [101]. In addition, to increase $L_{\max}$ and angle range, a transducer array can be adopted [102].

3.3.2. Measurement Accuracy of Ultrasonic Ranging System